Errata to "Unbounded field operators in categorical extensions of conformal nets"

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1. In Sec. 2.4, in the first paragraph of Def. 2.29, one should add the assumption that for each $\mathfrak{a}_1, \ldots, \mathfrak{a}_n \in \mathfrak{H}_i(\widetilde{I})$ and $c_1, \ldots, c_n \in \mathbb{C}$, the operators
$$c_1\mathcal{L}(\mathfrak{a}_1, \widetilde{I}) + \cdots + c_n\mathcal{L}(\mathfrak{a}_n, \widetilde{I}) : \mathcal{H}_k \to \mathcal{H}_i \boxdot \mathcal{H}_k$$
$$c_1\mathcal{R}(\mathfrak{a}_1, \widetilde{I}) + \cdots + c_n\mathcal{R}(\mathfrak{a}_n, \widetilde{I}) : \mathcal{H}_k \to \mathcal{H}_k \boxdot \mathcal{H}_i$$
are localizable; otherwise Rem. 2.30 may not hold.

2. In Sec. 2.7, Def. 2.42, there is a typo: $\widetilde{I}_2$ should be clockwise (rather than anticlockwise) to $\widetilde{I}_1$. Also, the second paragraph can be more precise. The revised version of Def. 2.42 (with only one paragraph) is as follows.

   Let $\mathscr{E}^w_{\mathrm{par}}$ be a categorical partial extension. Given $\mathcal{H}_{i_1}, \ldots, \mathcal{H}_{i_n} \in \mathcal{F}$, $\widetilde{I} \in \widetilde{\mathcal{J}}$, we define $\mathcal{H}_{i_n,\ldots,i_1}(\widetilde{I})$ to be the set of all
$$(\mathfrak{a}_n, \ldots, \mathfrak{a}_1) \in \mathfrak{H}_{i_n}(\widetilde{I}_n) \times \cdots \times \mathfrak{H}_{i_1}(\widetilde{I}_1)$$
such that $\widetilde{I}_t \subset \widetilde{I}$ and $\mathfrak{a}_t \in \mathfrak{H}_{i_t}(\widetilde{I}_t)$ for each $1 \leqslant t \leqslant n$, and $\widetilde{I}_s$ is clockwise to $\widetilde{I}_t$ whenever $1 \leqslant s < t \leqslant n$.

3. In the last paragraph of the proof of Thm. 3.39, the formula $\dim_{\mathcal{C}_V}(U) = \dim_{\mathrm{Rep}^0(A)}(U)$ is not correct, and needs to be deleted. In the subsequent sentence, the formula
$$D(\mathcal{C}_V)/D(\mathrm{Rep}^0(A)) = \dim_{\mathrm{Rep}^0(A)}(U)$$
citing [KO02] Thm. 4.5 is also citing incorrectly. The correct formula in that paper is
$$D(\mathcal{C}_V)/D(\mathrm{Rep}^0(A)) = \dim_{\mathcal{C}_V}(U)$$
These two mistakes indeed cancel each other, which again implies $D(\mathcal{C}_V)/D(\mathcal{C}_U) = \dim_{\mathcal{C}_V}(U)$ and hence finishing the proof of Thm. 3.39.

4. In the proof of Lem. 3.47, the last sentence "Thus they also hold for $V_\Lambda \otimes V_L$ and hence for V_Λ by theorems 3.38 and 3.39" is a little misleading. The more appropriate one should be "Thus they also hold for $V_\Lambda \otimes V_L$ (by theorems 3.38 and 3.39) and hence for V_Λ ".

5. In the beginning paragraph of Ch. A, the statement "$\overline{p_s A^* A} = p_s A^* A p_s$ is bounded and positive" should be "$\overline{q_s A^* A} = q_s A^* A q_s$ is bounded and positive".